

deviation. For statistical analysis, the Mann-Whitney and Wilcoxon tests were applied to compare PD and CG and to compare the means of the body composition values generated by the two devices. Spearman's correlation test was used to measure the association between the variables. The level of statistical significance applied was $p < 0.05$.

Results: The sample consists of 19 men (32%) and 11 women (18%) in the PD group and 19 men and 11 women in the CG. In the GC versus PD group analysis by the BIS, the PD group presented lower weight (68 ± 13 kg versus 86 ± 16 kg), older age (52 ± 1.9 years versus 37 ± 1.9 years), lower body mass index (24 ± 3.4 kg/m² versus 29 ± 4.7 kg/m²), higher water overload ($+2.1 \pm 2.2$ L versus -3.2 ± 9.5 L), lower intracellular water/total body water index (51 ± 4.6 versus 57 ± 2.1), higher extracellular water/total body water index (49 ± 4.2 versus 43 ± 2.1) and lower phase angle ($4.7 \pm 1.1^\circ$ versus $6.9 \pm 0.72^\circ$). In the BIA versus BIS analysis of comparison within the PD group and within the CG, the means of total body water (L), fat-free mass (kg and %) and body cell mass (kg) were higher by BIA and the means of fat mass (kg and %) were higher by BIS in both groups. The correlation coefficients (r) between the measurements of the two devices were significant for all body composition data and weaker within the PD, except for total body water.

Conclusion: For the analyzed sample, with regard to body composition and compared to CG, the PD group was older, with lower body mass index, lower phase angle and more water overload, especially in the extracellular compartment, which is expected due to its clinical condition. The weaker correlations between device measurements on PD group are possibly caused by the abnormal water distribution, a limiting factor for bioelectrical impedance analysis (of both devices), which may overestimate more hydrated body tissues as fat free mass.

Disclosure of Interest: None declared

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ASSESSMENT OF ADIPOSITY IN CHRONIC KIDNEY DISEASE.

N.T. Bellafronte^{*1}, G.B. Cuadrado², P.G. Chiarello¹. ¹Health Sciences Department, Ribeirão Preto Faculty of Medicine, University of São Paulo, Ribeirão Preto, Brazil; ²Nephrology Department, Hospital Universitario La Princesa, Universidad Autonoma de Madrid, Madrid, Spain

Rationale: The presence of muscle depletion and increased adiposity are common in chronic kidney disease (CKD) as a result of the higher protein degradation, resistance of protein synthesis, side effects of treatment applied, low physical activity, diet restriction and life style. Bioelectrical impedance analysis is a low cost, bedside, portable and easy to use in clinical routine method for nutritional assessment. Specific bioelectrical impedance vector analysis (BIVAs) is a vectorial analysis of bioelectrical impedance data of body composition that have been poorly evaluated, mainly in CKD. We aim to evaluate the application of BIVAs for nutritional status assessment in CKD.

Methods: We evaluated CKD patients on non-dialysis-dependent treatment (NDD), peritoneal dialysis (PD), hemodialysis (HD) and kidney transplant (KTx). Body composition was assessed by Dual-energy X-ray absorptiometry with total and appendicular lean mass, total and central fat mass data. Muscle strength was evaluated by handgrip strength and phase angle was obtained by bioelectrical impedance analysis. Nutritional status was classified by BIVAs into high and low lean mass, high and low fat percentage. Data is presented as mean, standard deviation and percentage. We applied the ANOVA test with post Bonferroni test for comparison between groups classified by BIVAs ($p < 0.05$).

Results: We evaluated 266 patients aged 47 ± 10 years. Of the 137 men, 15 were classified as high and 118 as low lean mass, 42 as high and 91 as low percentage of fat, by BIVAs. Of the 129 women, 23 were classified as high and 104 as low lean mass, 56 as high and 70 as low percentage of fat. Of the 83 NDD patients, 14 were classified as high and 66 as low lean mass, 2 as high and 50 as low fat percentage. Of the 79 patients on HD, 12 were classified as high and 63 as low lean mass, 34 as high and 42 as low percentage of fat. Of the 23 patients on PD, 4 were classified as high and 18 as low lean mass, 2 as high and 20 as low percentage of fat. Of the 81 patients on KTx, 8 were classified as high and 72 as low lean mass, 32 as high and 48 as low fat percentage. Patients classified as having high and low lean mass, high and low fat percentage presented, respectively: weight (kg), 99 ± 46 , 87 ± 46 , 74 ± 15 and 67 ± 15 ; body mass

index (kg/m²), $26 \pm 4.26 \pm 5$, 30 ± 4.5 and 24 ± 3.9 ; hand grip strength (kg), 35 ± 3.5 , 36 ± 3.9 , 29 ± 11 and 32 ± 12 ; phase angle ($^\circ$), 7 ± 0.5 , 5.7 ± 0.9 , $6 \pm 0.9^\circ$ and 5.6 ± 1 ; lean mass (kg), $39 \pm 9.41 \pm 10$, 41 ± 10 and 40 ± 10 ; lean mass index (kg/m²), 14.8 ± 2.3 , 15.1 ± 2.7 , 16 ± 4.2 and 14.6 ± 2.9 ; appendicular lean mass index (kg/m²), 6.8 ± 1.2 , 6.8 ± 1.4 , 7.2 ± 2.1 and 6.6 ± 1.5 ; fat mass (kg), 20 ± 7 , 23 ± 8 , 28 ± 7 and 20 ± 7 ; fat mass index (kg/m²), 7.9 ± 3.1 , 8.8 ± 3.4 , 11.2 ± 3.2 and 7.2 ± 2.7 ; trunk fat mass (kg), 13 ± 10 , $13 \pm 7.15 \pm 4$ and 12 ± 9 . Weight, phase angle, total and appendicular lean mass, fat mass and fat mass index were different between the groups: high lean mass versus high percentage of fat mass, low lean mass versus high percentage of fat mass, low lean mass versus low percentage of mass fat and high versus low percentage of fat mass. There was no difference between the groups high versus low lean mass, high lean mass versus low percentage of fat mass.

Conclusion: Both sex and all CKD treatment groups showed important impairment of nutritional status according to BIVAs. BIVAs was able to distinguish between patients with different adiposity. However, BIVAs was not able to distinguish between individuals with different muscle mass status. Therefore, in patients with CKD, BIVAs could be adequate for assessing nutritional status in terms of adiposity.

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THE RELATIONSHIP OF ADHERENCE TO THE MEDITERRANEAN DIET WITH DISEASE ACTIVITY AND QUALITY OF LIFE IN CROHN'S DISEASE PATIENTS.

N.D. Gkogkou^{*1}, I. Migdanis^{1,2}, K. Oikonomou¹, C. Valaris³, A. Migdanis^{1,2}, E. Armeni², A. Kapsoritakis¹, S. Potamianos¹. ¹Department of Gastroenterology, University of Thessaly, Larissa; ²Department of Nutrition and Dietetics, University of Thessaly, Trikala; ³School of Pharmacy, Aristotle University, Thessaloniki, Greece

Rationale: A specific nutritional intervention could be a promising way of managing Crohn's disease (CD) flare ups. Mediterranean diet (MD) might be a likely candidate due to its beneficial properties which could offer multiple benefits to CD patients. The purpose of the present study was to investigate the relationship between the adherence to the MD and the degree of disease activity, as well as the quality of life in patients with CD.

Methods: In the present prospective observational study patients diagnosed with CD were included for evaluation. Medical history and anthropometric measurements were recorded. Disease activity was assessed using the Harvey & Bradshaw Index (HBI). Adherence to the MD was assessed using the MedDiet score. Furthermore, the participants' quality of life was assessed using the Crohn's and ulcerative colitis questionnaire (CUCQ-8). Interrelations between MedDiet score and HBI and CUCQ-8 were evaluated.

Results: In total 60 patients (63.3% females) with a median age of 36.5 (range 27–46.8) years were evaluated. According to the HBI score 28 (46.7%) patients had active disease ($5 \leq \text{HBI} \leq 14$) while 32 (53.3%) patients were in remission ($\text{HBI} \leq 4$). Comparing patients in remission with those who had active disease, results showed that patients in remission were more attached to the MD compared to patients with active disease who were less attached ($p = 0.02$). Moreover, patients with increased adherence to the MD (MedDiet score 28–55) had a higher score in quality of life assessment compared to patients with a lower adherence degree to the MD (MedDiet score 0–27) ($p = 0.05$).

Conclusion: In conclusion, higher adherence to the MD showed a beneficial effect on CD patients' clinical condition, by being correlated with reduced disease activity as well as by being associated with improved quality of life.

Disclosure of Interest: None declared

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UTILITY OF A MODIFIED CONTROLLING NUTRITIONAL STATUS SCORE IN MORTALITY RISK ASSESSMENT OF PATIENTS ON HEMODIALYSIS

N. Morimoto^{*1,2,3}, K.Y. Urayama³, K. Ozawa¹, H. Tanaka². ¹Yokosuka Clinic; ²Department of Nephrology, Yokosuka Kyosai Hospital, Yokosuka;